

COMPUTER GRAPHICS (CAC305) LECTURE 24

Er Sanim Kumar Khatri

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Animation

What is Animation?

Animation is the process of creating the illusion of motion by displaying a series of static images (frames) in rapid succession. Each frame is slightly different from the previous one, and when played back at sufficient speed (typically 24–30 frames per second), the human eye perceives smooth, continuous movement. This phenomenon is known as **persistence of vision**.

In computer animation, movement is generated by software that renders and sequences images digitally, rather than drawing each frame by hand.

Animation

Types of Animation

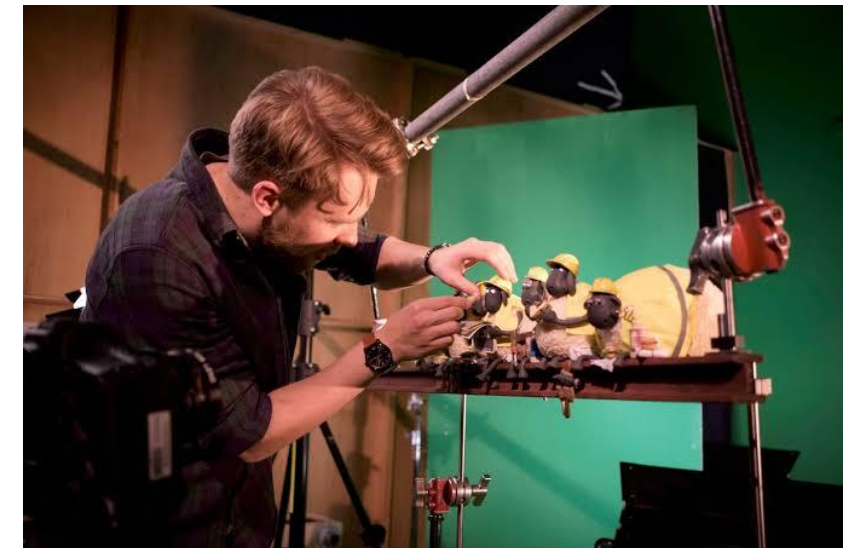
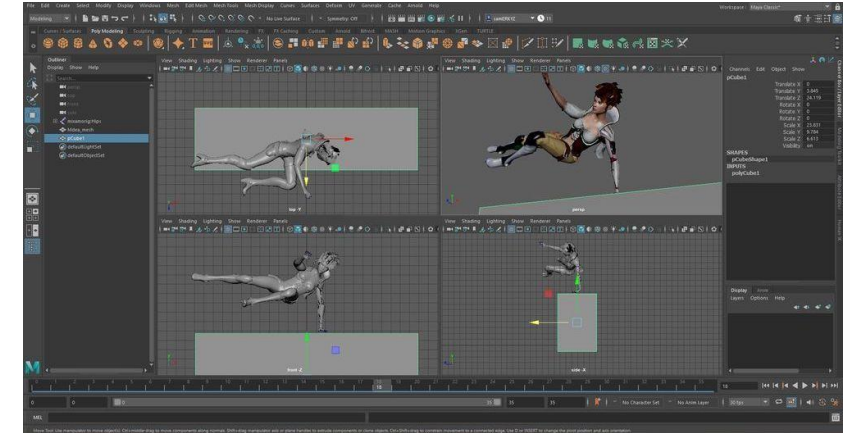
1. **Traditional Animation (Cel Animation)** The oldest form of animation, where each frame is hand-drawn on paper or transparent celluloid sheets (cels). Frames are photographed in sequence. Classic Disney films like *Snow White* (1937) used this technique. It is labor-intensive and largely replaced by digital tools today.
2. **2D Animation** Digital 2D animation uses software to draw and animate in a two-dimensional space. Tools like Adobe Animate (formerly Flash), Toon Boom Harmony, and After Effects are commonly used. It can replicate the look of traditional animation or create entirely new visual styles.



Animation

Types of Animation

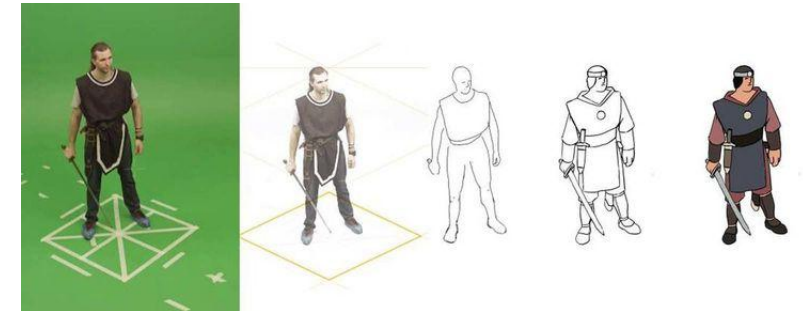
- 3. 3D Animation (CGI)** Computer-generated imagery (CGI) is used to create characters and environments in a three-dimensional virtual space. Animators rig 3D models and pose them across keyframes; software calculates the in-betweens. Used extensively in films (Pixar, DreamWorks), video games, and VFX.
- 4. Stop Motion Animation** Physical objects such as puppets, clay figures, cut-outs, or even real-world objects, are photographed one frame at a time, with small adjustments made between shots. When played back, they appear to move. Examples include *Shaun the Sheep* and *Kubo and the Two Strings*.



Animation

Types of Animation

5. **Motion Graphics** Animated graphic design, typically used for text, logos, and abstract shapes in titles, infographics, advertisements, and broadcast design. Adobe After Effects is the industry standard tool.
6. **Rotoscoping** An animation technique where animators trace over real, live-action footage frame by frame to produce realistic movement. This was used in classic Disney films and is still used today for reference and visual effects.

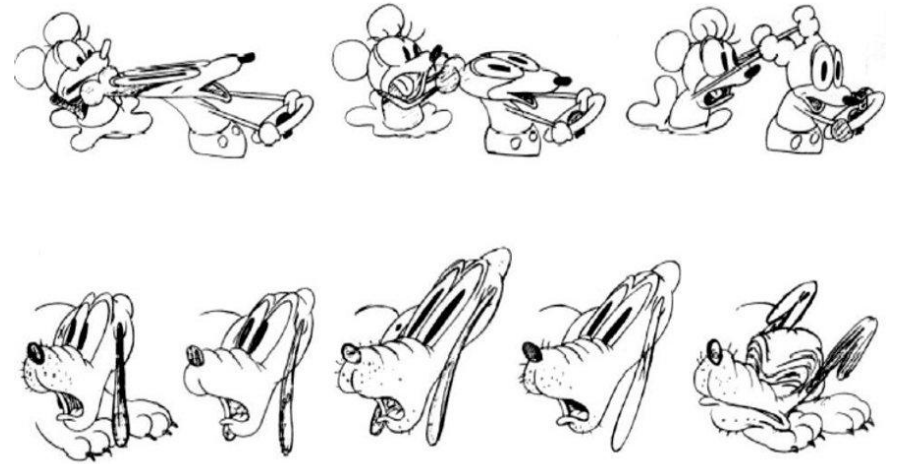


Animation

The 12 Basic Principles of Animation

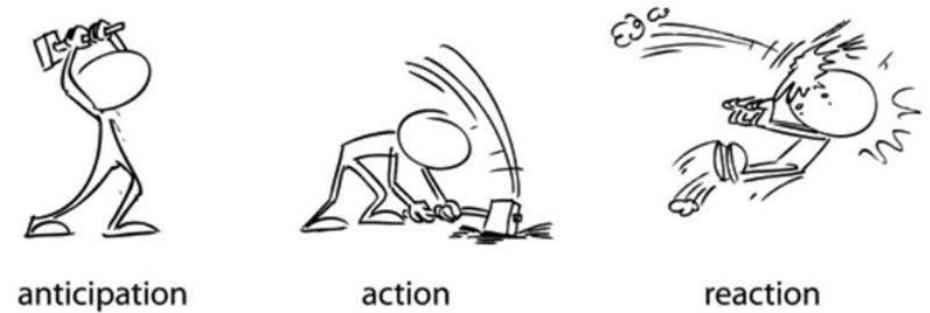
These principles were developed by Disney animators Ollie Johnston and Frank Thomas and were first formally outlined in their 1981 book *The Illusion of Life: Disney Animation*. They remain the foundational guidelines for creating believable and effective animation.

- 1. Squash and Stretch** This principle conveys the physical property of an object changing shape during motion. When a ball drops and hits the ground, it flattens (squash); as it bounces back up, it elongates (stretch). This makes animation feel alive and believable. The volume of the object must always remain constant, squash and stretch only changes the shape, not the mass.

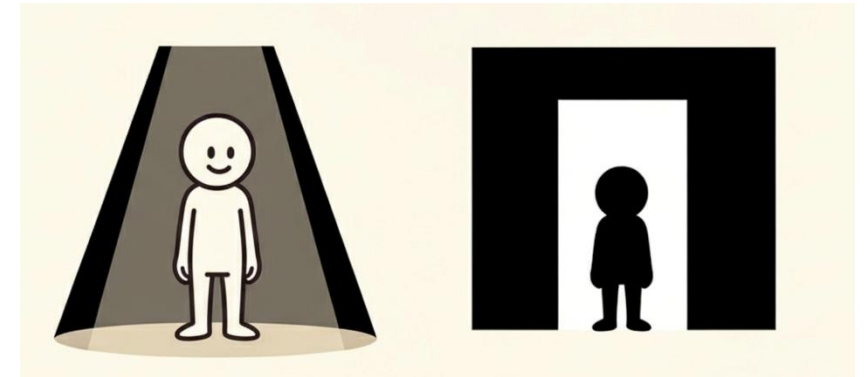


Animation

2. **Anticipation:** An action typically has three parts: the preparation (anticipation), the action itself, and the reaction. Anticipation prepares the viewer for what is about to happen. For example, a character bends their knees and lowers their body before jumping. Without anticipation, movements appear sudden and unnatural.



3. **Staging:** Staging is the art of presenting an idea or action clearly so the audience understands it. It involves controlling the placement of characters, camera angles, lighting, and background to direct the viewer's attention to what matters most in a scene.

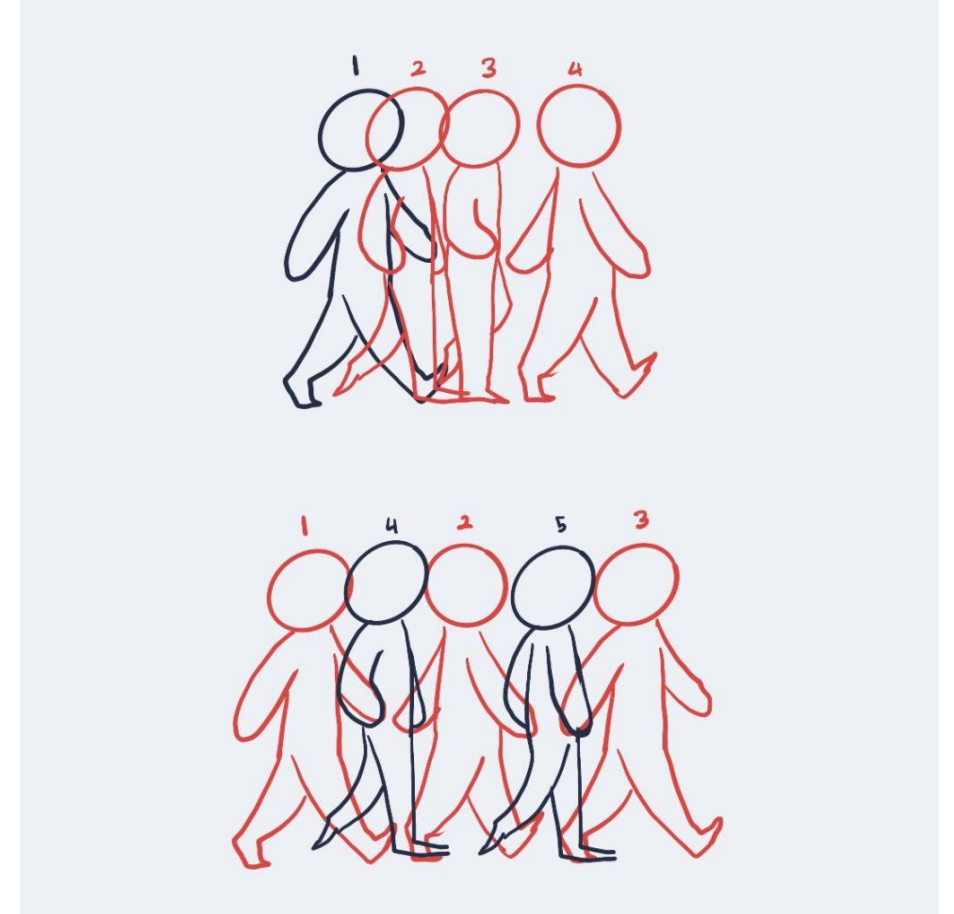


Animation

4. **Straight Ahead Action and Pose to Pose:** These are two distinct approaches to animating:

- **Straight Ahead:** The animator draws each frame sequentially from beginning to end. This produces fluid and spontaneous motion but can cause proportions to drift.
- **Pose to Pose:** The animator draws the key poses (start and end), then fills in the in-between frames. This gives better control over timing and is preferred for scenes requiring precision and acting.

In computer animation, pose-to-pose is common since software handles the in-betweens automatically.



Animation

5. Follow Through and Overlapping Action:

Follow Through: Parts of a character or object continue to move after the main body has stopped (e.g., a dog's tail still wagging after it sits).

Overlapping Action: Different parts of a body move at different rates simultaneously. Not everything moves together, hair, clothing, and limbs all react at slightly different times.

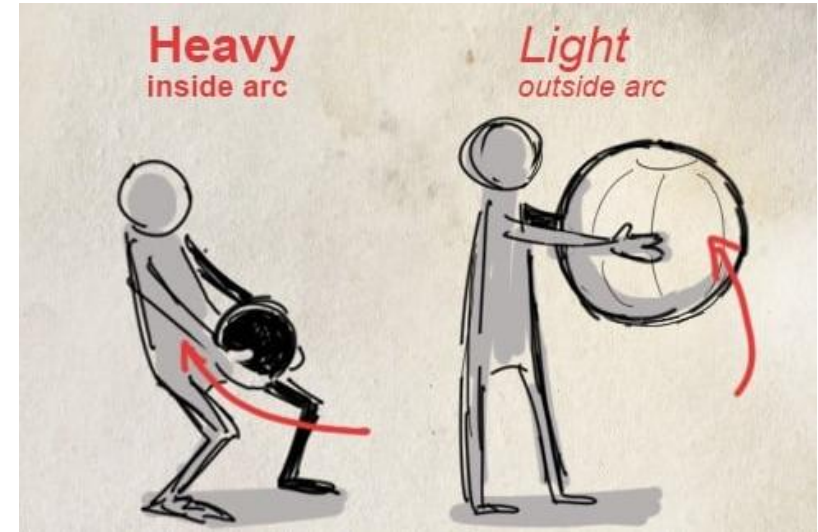
6. Slow In and Slow Out (Ease In and Ease Out)

Objects in the real world rarely start or stop abruptly. They accelerate when starting (slow in) and decelerate when stopping (slow out). Adding more frames near the beginning and end of a motion, and fewer in the middle, simulates this natural effect.



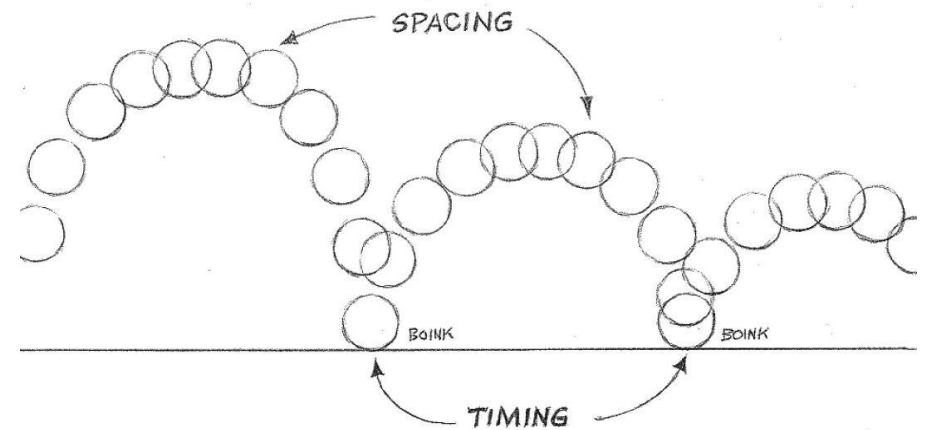
Animation

- 7. Arcs:** In nature, most living creatures move along curved paths rather than straight lines. A swinging arm, a jumping character, or a bouncing ball all follow arc trajectories. Using arcs in animation creates more natural-looking movement.
- 8. Secondary Action:** Secondary actions reinforce the main action and add dimension to a character. For example, as a character walks (main action), they might swing their arms or bob their head (secondary actions). Secondary actions must complement, not compete with, the main action.



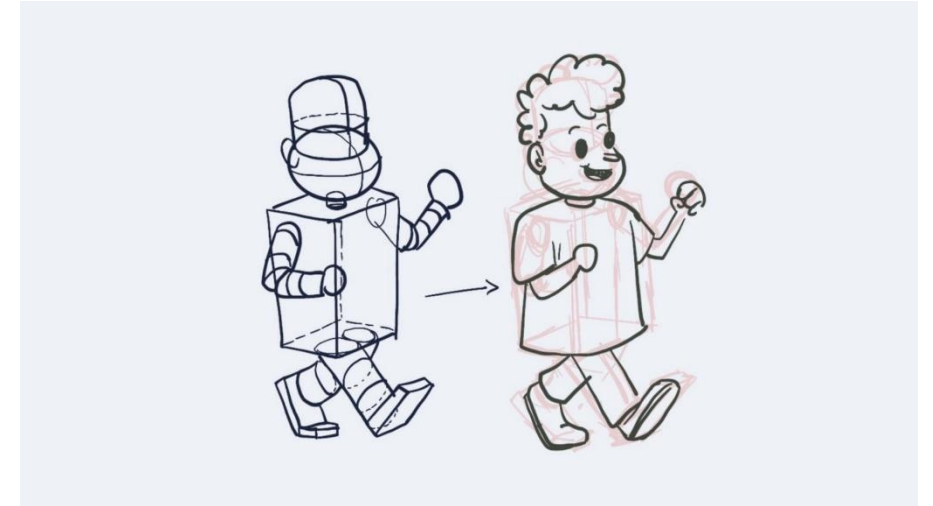
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9. **Timing:** Timing refers to the number of frames used for an action and directly affects how the animation feels. More frames = slower, heavier movement. Fewer frames = faster, snappier movement. Timing is what gives animation weight, emotion, and personality.
10. **Exaggeration** To create a visually appealing and engaging animation, movements and expressions are often pushed beyond reality. A surprised character's eyes might pop out of their head. This exaggeration does not mean being unrealistic — rather, it means amplifying reality to make emotions and movements read clearly.



Animation

11. Solid Drawing: Even in digital animation, the animator must understand form, weight, volume, and 3D space. Characters and objects should look three-dimensional and consistent from all angles. Poor draftsmanship leads to flat, unconvincing characters.



12. Appeal: Appeal in animation is similar to charisma in real life. Characters must be interesting and pleasing to look at, whether they are heroes or villains. A well-designed character with clear readable expressions and appealing proportions draws the audience in.



Thank you

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What is Adobe Flash / Adobe Animate?

Adobe Flash (now officially called Adobe Animate) is a 2D animation and multimedia authoring software. It was first launched in 1996 and became the dominant tool for web animation, interactive content, and web-based games for over a decade. The animations created in Flash are vector-based, meaning they scale to any resolution without losing quality, unlike pixel/raster animations.

Flash produces output in formats such as **.swf** (Shockwave Flash) and **.fla** (source file). Today, Adobe Animate supports export to HTML5 Canvas, WebGL, and video formats.

Key Components of the Flash Interface

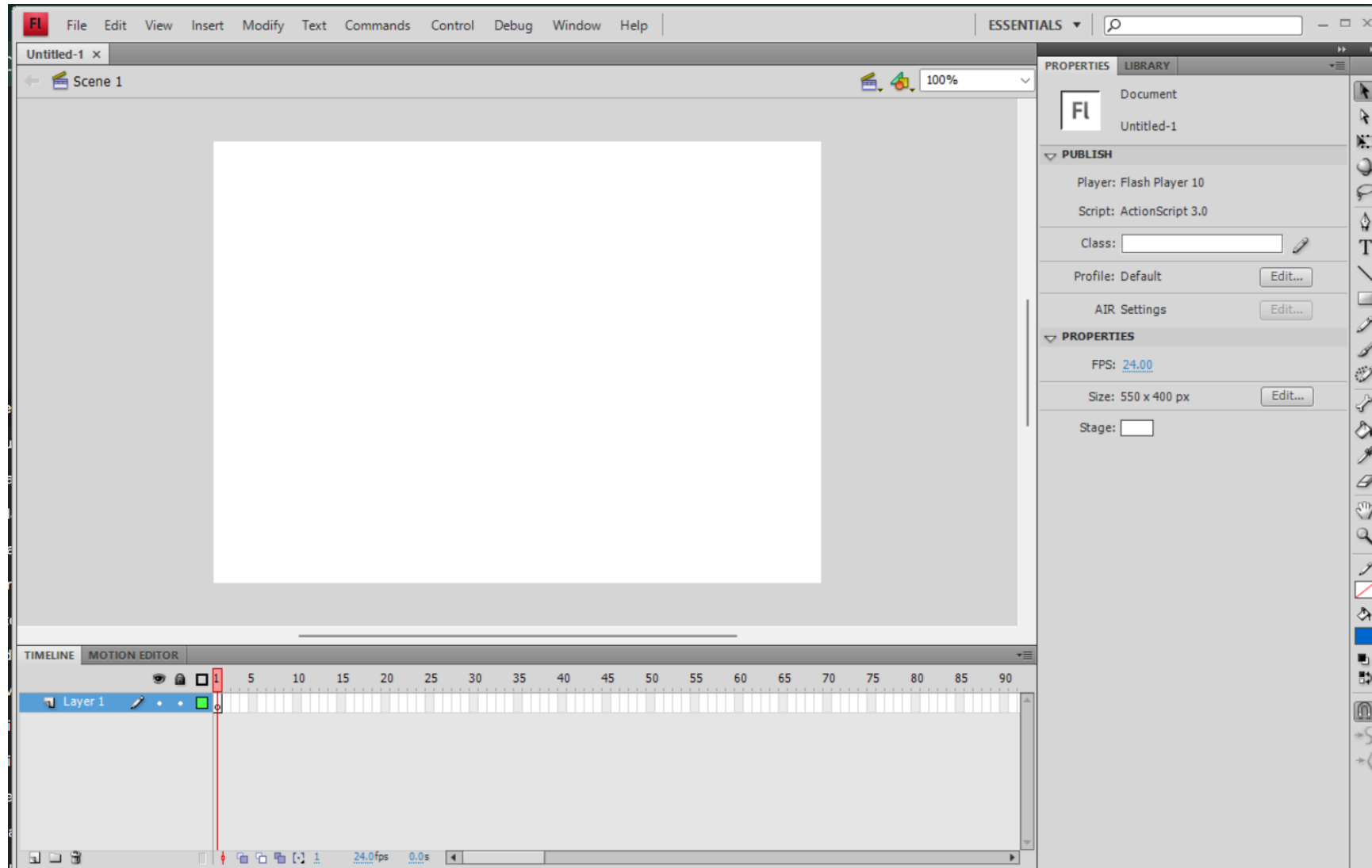
1. **Stage:** The Stage is the central white rectangular area where you compose and view your animation. It represents what the final audience will see. Objects placed outside the Stage boundaries are not visible in the final output but can be animated to move onto or off the Stage.
2. **Timeline:** The Timeline, located at the top or bottom of the workspace, controls the sequence and timing of animation. It is organized into:
 1. Frames: Each individual unit of time.
 2. Keyframes: Special frames where content changes occur (marked by a black dot).
 3. Blank Keyframes: Empty keyframes that indicate no content on the Stage at that point.
 4. Layers: Horizontal rows in the Timeline that allow independent control of different elements.

Adobe flash

Key Components of the Flash Interface

3. **Tools Panel (Toolbox):** Located on the left side, the Tools Panel contains all the drawing and selection tools needed to create and manipulate objects.
4. **Properties Panel (Property Inspector):** The Properties Panel appears at the bottom or side of the workspace and dynamically shows the properties of whatever is currently selected: a shape, symbol, frame, or the Stage itself. Here you can change stroke color, fill color, width, height, coordinates, instance names, and more.
5. **Library Panel:** The Library stores all reusable assets in your Flash document — symbols (movie clips, graphics, buttons), imported bitmaps, sounds, and videos. Once created or imported, an asset can be dragged from the Library onto the Stage multiple times without increasing the file size significantly.
6. **Menu Bar:** The Menu Bar at the top provides access to all commands: File (new, open, save, publish), Edit (cut, copy, paste, undo), View (zoom, rulers, guides), Insert (new symbol, keyframe), Modify (shape, transform), Control (play, test movie), and more.

Adobe flash



Setting Stage Dimensions, Working with Panels, Panel Layouts

Setting Stage Dimensions

The Stage dimensions define the canvas size of your Flash animation, what the audience sees. To set or change the Stage dimensions:

- Go to **Modify** → **Document** (or press Ctrl+J).
- The **Document Properties** dialog opens, where you can set:
 - **Width** and **Height** in pixels (e.g., 550 × 400 px is a traditional web default).
 - **Background Color** of the Stage.
 - **Frame Rate** (fps — frames per second). Common values: 12 fps for basic web animation, 24 fps for smoother animation, 30 fps for video quality.
 - **Ruler Units** (pixels, centimeters, inches).
- Click **OK** to apply.

Adobe flash

Standard Stage size include:

Purpose	Dimensions
Web (traditional)	550 × 400 px
HD Video	1920 × 1080 px
Mobile	320 × 480 px
Square (social media)	1080 × 1080 px

Working with Panels

Panels in Flash are floating or docked windows that give you access to specific tools, settings, and properties. The most commonly used panels include:

- **Timeline Panel:** controls frame sequence and layer order.
- **Properties Panel:** shows properties of selected objects or the current frame.
- **Library Panel:** stores and manages reusable assets.
- **Color Panel:** creates and edits colors and gradients.
- **Align Panel:** aligns multiple objects relative to each other or the Stage.
- **Transform Panel:** applies precise numerical transformations (scale, rotate, skew).
- **Actions Panel (F9):** used to write ActionScript code for interactivity.
- **Info Panel:** shows size and position of selected objects in real time.
- **Opening/Closing Panels:** Panels can be opened from the **Window** menu. Panels can be toggled on or off as needed.

Adobe flash

Panel Layouts

Panel layout refers to the arrangement of panels in the workspace. Flash provides several pre-configured workspace layouts accessible from **Window** → **Workspace**:

- **Essentials:** a clean layout for beginners with the most important panels visible.
- **Animator:** optimized for animation work with Timeline prominent.
- **Designer:** for graphic design tasks.
- **Classic:** the traditional Flash layout used in older versions.
- **Custom:** users can arrange panels by dragging, docking, or floating them, then save the layout with a custom name.

Working with Panel Layouts:

Panels can be docked to any side of the screen by dragging them to the edge.

Panel groups can be collapsed to icons to save screen space.

Use **Window** → **Workspace** → **Reset [workspace name]** to restore the default layout.

Multiple monitors benefit from spreading panels across screens for more Stage workspace.

Thank you

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Layers and Views

Understanding Layers

Layers in Flash work like transparent sheets stacked on top of one another. Objects on higher layers appear in front of objects on lower layers. Layers are managed in the **Timeline panel**, each represented as a horizontal row.

Why use Layers?

- To organize different elements (background, characters, text, sounds) separately.
- To animate objects independently without affecting other elements.
- To apply special effects like masks.
- To keep a logical, editable structure in complex animations.

Types of Layers

- 1. Normal Layer** The default layer type. Used for placing and animating objects on the Stage.
- 2. Guide Layer** A guide layer helps you align objects precisely. Objects on a guide layer are visible while editing but do not appear in the final published output. Used with rulers and grid lines for positioning reference.
- 3. Motion Guide Layer** A special guide layer used in **classic tween** animation. You draw a path on the motion guide layer, and the animated object follows that path (e.g., a butterfly flying along a curved route). The guide layer is linked to (sits above) the layer containing the animated object.
- 4. Mask Layer** A mask layer works like a stencil. The shape on the mask layer defines which parts of the layer(s) below it are visible, only areas where the mask shape overlaps are revealed; everything else is hidden.
- 5. Locked Layer** Any layer can be locked to prevent accidental edits. A locked layer is shown with a padlock icon and cannot be modified until unlocked.

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Views in Flash

Flash provides several view options to help you work more effectively.

1. Normal View The standard view showing objects in full color on the Stage.

2. Outline View Objects on a layer are displayed as colored outlines only, making it easier to see overlapping shapes and paths. Each layer's outlines use a unique color.

3. Fit in Window / Zoom

- **Ctrl+Shift+Alt+F** : fits the entire Stage in the window.
- **Zoom In (Ctrl++) / Zoom Out (Ctrl+-)**: adjusts the view magnification.
- **Zoom Tool (Z)**: click on the Stage to zoom in; Alt+click to zoom out.

4. Show Grid View → Grid → Show Grid displays a grid over the Stage to help align objects. The grid spacing can be customized.

5. Show Rulers View → Rulers displays measurement rulers along the top and left edge of the Stage window. Useful for precise positioning.

6. Snapping Snapping causes objects to automatically align to the grid, guides, or other objects when moved close to them. Toggle with **View → Snapping**.

Adobe flash

Shaping Objects: Overview of Shapes, Drawing and Modifying Shapes

Overview of Shapes in Flash

In Adobe Flash/Animate, there are two fundamental types of graphic objects:

1. **Shapes (Drawing Objects / Merge Drawing)** Raw vector shapes drawn directly on the Stage. When two shapes on the same layer overlap, they interact: the shape on top can cut into or merge with the shape below. This is the default drawing mode in Flash.
2. **Drawing Objects (Object Drawing Mode)** When Object Drawing Mode (toggle with J) is active, shapes are drawn as self-contained objects that do not merge with other shapes. They behave more like individual entities and are recommended for most animation work to prevent accidental merging.
3. **Symbols:** Symbols are reusable objects stored in the Library. There are three types:
 - **Graphic Symbol:** used for static images or simple animations tied to the main Timeline.
 - **Movie Clip Symbol:** has its own independent Timeline; can be animated independently of the main Timeline.
 - **Button Symbol:** a 4-frame symbol used for interactive click/hover/press states.

Converting a shape to a symbol (F8) is required before applying motion tweens.

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Bitmap Images and Sounds

Bitmap Images in Flash

A **bitmap image** (also called a raster image) is made up of a grid of pixels, where each pixel has a specific color. Unlike Flash's native vector graphics, bitmaps are resolution-dependent, they can appear blurry or pixelated when scaled up. Common bitmap formats include JPEG, PNG, GIF, and BMP.

Importing Bitmaps

Go to **File** → **Import** → **Import to Stage** (places directly on the Stage) or **Import to Library** (stores in the Library for later use).

The imported image appears in the Library panel.

Drag it from the Library onto the Stage as needed.

Using Bitmaps in Flash

A bitmap can be used as a background, texture, or sprite.

You can place multiple instances of the same bitmap from the Library without increasing file size proportionally.

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Converting Bitmap to Vector (Trace Bitmap)

Flash can convert a bitmap into an editable vector graphic using **Modify → Bitmap → Trace Bitmap**.

Options include:

- **Color Threshold:** controls how similar colors are grouped together.
- **Minimum Area:** the smallest region of color to include.
- **Curve Fit:** controls how smoothly curves are traced.
- **Corner Threshold:** controls how sharp corners are reproduced.
- The result is an editable vector that can be resized without loss of quality, but it may not look identical to the original bitmap.
- **Breaking Apart a Bitmap** Selecting a bitmap and pressing **Ctrl+B** (Modify → Break Apart) converts it into a fill, allowing you to paint over it or select color areas with the eyedropper and paint bucket tools.

The result is an editable vector that can be resized without loss of quality, but it may not look identical to the original bitmap.

Breaking Apart a Bitmap Selecting a bitmap and pressing **Ctrl+B** (Modify → Break Apart) converts it into a fill, allowing you to paint over it or select color areas with the eyedropper and paint bucket tools.

Adobe flash

Working with Sounds in Flash

Flash supports several audio file formats for import, including **MP3**, **WAV**, and **AIFF**.

Importing Sounds

- Go to **File** → **Import** → **Import to Library**.
- The audio file is added to the Library and can be used in the Timeline.

Adding Sound to a Timeline

- Select the frame in the Timeline where you want the sound to begin.
- Open the **Properties Panel**; under the Sound section, select the audio file from the Name dropdown.
- Choose the **Sync** setting:

Thank you

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Animation

Animation in Flash: Core Concepts

Before animating, it's important to understand a few core concepts:

- Frame: A single image in the animation sequence.
- Keyframe: A frame where a change occurs (new content, new position, new state). Indicated by a filled black circle (●) in the Timeline.
- Blank Keyframe: An empty keyframe with no content, indicated by an empty circle (○).
- Frame Rate (fps): How many frames play per second. Default in Flash is 24 fps.
- Playhead: The red marker in the Timeline indicating the current frame being displayed.

Frame by Frame

Frame-by-Frame Animation

In frame-by-frame animation, the content of each individual frame is manually created or altered. This is the most labor-intensive form of animation but gives the animator full control over every drawing.

How it works:

- Insert a keyframe at each frame where content changes (F6 to insert a keyframe).
- Draw or modify the content on the Stage at each keyframe.
- When the animation is played back, the eye perceives the sequence of images as continuous motion.

Steps to Create Frame-by-Frame Animation:

- Select frame 1 in the Timeline; draw your object on the Stage.
- Press F6 to insert a new keyframe in frame 2.
- Modify or redraw the object slightly (e.g., move a character's arm).
- Continue pressing F6 and modifying content for each subsequent frame.
- Press Enter to preview.

Frame by Frame

Advantages:

- Full creative control over every drawing.
- Can achieve hand-crafted, organic look.
- Suitable for complex character animation.

Disadvantages:

- Time-consuming and labor-intensive.
- Larger file sizes (each frame is a separate image).
- Difficult to edit retroactively.

Onion Skinning:

Flash provides an Onion Skin feature (button in the Timeline) that shows semi-transparent versions of frames before and after the current frame, helping animators trace and align drawings consistently across frames.

Tweening

Tweening

Tweening (short for in-betweening) is the process of automatically generating intermediate frames between two defined keyframes. The animator sets the start state and the end state; Flash calculates everything in between. This dramatically speeds up the animation workflow.

There are three types of tweens in Flash/Animate:

1. Motion Tween
2. Shape Tween
3. Classic Tween

Tweening

1. Motion Tween

Motion Tween animates the position, size, rotation, color, and opacity of a **symbol** across frames.

Requirements:

- The object must be a **Symbol** (Movie Clip, Graphic, or Button).
- One object per tween span.

Steps to Create a Motion Tween:

- Place a symbol on the Stage in frame 1.
- Right-click on the frame in the Timeline → **Create Motion Tween**.
- Move the playhead to a later frame (e.g., frame 24).
- Drag the symbol to a new position on the Stage. A keyframe is automatically created.
- Flash generates a smooth motion between the two positions.
- Press **Enter** to preview.

Tweening

Properties that can be tweened:

- Position (X, Y)
- Scale (width, height)
- Rotation
- Alpha (opacity/transparency)
- Color tint
- Filters (blur, shadow, etc.)

Motion Path: Flash displays the motion path as a colored line on the Stage. This path can be edited using the Selection Tool to curve it, creating arced or curved movements.

Easing: Easing controls the acceleration at the start and end of a tween.

- **Ease In** (negative value): Object accelerates — starts slow, ends fast.
- **Ease Out** (positive value): Object decelerates — starts fast, ends slow.

Tweening

2. Shape Tween

Shape Tween is used to **morph** one vector shape into another. It works on raw vector drawings, not symbols.

Requirements:

- The object must be a **raw shape** (not a symbol, not a group).
- Both keyframes must contain shapes.

Steps to Create a Shape Tween:

- Draw a vector shape on the Stage in frame 1.
- Insert a blank keyframe (**F7**) at a later frame.
- Draw a different shape (or modify the existing shape) in this second keyframe.
- Click anywhere between the two keyframes.
- In the **Properties Panel**, set the Tween type to **Shape**.
- The Timeline fills with a light green background with an arrow, indicating a valid shape tween.
- Press **Enter** to preview the morphing effect.

Tweening

Shape Hints: If Flash morphs incorrectly (shapes twisting or distorting in unexpected ways), you can use Modify → Shape → Add Shape Hint to manually tell Flash which point on the starting shape corresponds to which point on the ending shape. Shape hints are labeled a, b, c, etc.

Difference Between Motion and Shape Tween:

Feature	Motion Tween	Shape Tween
Works on	Symbols	Raw shapes
Effect	Moves/transforms objects	Morphs shapes
Timeline color	Blue/purple	Green
File size	Smaller	Larger

Tweening

3. Classic Tween

Classic Tween is the older style of motion tween available in earlier versions of Flash. It operates similarly to Motion Tween but uses a different workflow:

- Requires manually placing keyframes at both the start and end of the animation.
- Allows use of Motion Guide Layers for path-based animation.
- Properties like alpha, color, and rotation can be changed between the two keyframes.

Classic Tweens appear in the Timeline with a **blue background** and an arrow. A broken arrow (dashed line) indicates an error in the tween setup.

Motion Guide with Classic Tween:

- Right-click the layer containing the classic tween → **Add Classic Motion Guide.**
- Draw a path on the guide layer using the Pencil Tool.
- Snap the symbol's center to the start of the path in the first keyframe.
- Snap the symbol's center to the end of the path in the last keyframe.
- Check **Orient to Path** in Properties to make the object face the direction of travel.

Masks

A **mask** in Flash works like a stencil or cutout. The shape placed on the mask layer defines what is visible in the layer(s) below it. Only the areas where the mask shape overlaps with the content below are revealed to the viewer; everything else is hidden.

Creating a Mask:

- Create the content you want to be partially revealed on one layer (e.g., Layer 1).
- Create a new layer directly above Layer 1 (e.g., Layer 2).
- On Layer 2, draw the shape that will act as the mask (this defines the visible area).
- Right-click on Layer 2 → **Mask**.
- Both layers lock automatically and the mask effect is applied.

Animated Masks: Masks become especially powerful when animated. You can animate the mask shape (using any tween type) to create effects like:

- **Spotlight effects:** A circular mask moves across a dark scene, simulating a moving spotlight.
- **Wipe transitions:** A rectangle grows from left to right to reveal text or images.
- **Rotating reveals:** A mask shape rotates to uncover content.
- **Text reveals:** Text on a lower layer is gradually revealed by a moving shape on the mask layer.

Masks

Notes on Masks:

- A single mask layer can mask multiple layers (drag layers beneath the mask layer and indent them).
- Gradients and images on a mask layer do NOT create soft edges — masks in Flash are binary (on/off). For soft edges, use external techniques.
- To edit the mask, click the padlock icon on the mask layer to temporarily unlock it.

Thank you